РОССИЙСКИЙ УНИВЕРСИТЕТ ДРУЖБЫ НАРОДОВ

Факультет физико-математических и естественных наук

ОТЧЁТ

ПО ЛАБОРАТОРНОЙ РАБОТЕ №3

дисциплина: Администрирование сетевых подсистем

Студент: Талебу Тенке Франк Устон

Студ. билет № 1032224534

Группа: НФИбд-02-23

МОСКВА

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Цель работы:

Целью данной работы является приобретение практических навыков по установке и конфигурированию DHCP-сервера.

Выполнение работы:

```
root@server:~

[vagrant@server ~]$ sudo -i
[root@server ~]# dnf -y install kea
Last metadata expiration check: 0:16:46 ago on Wed 04 Mar 2026 03:29:26 PM UTC.
Dependencies resolved.
================================================================================
 Package                       Arch      Version             Repository    Size
================================================================================
Installing:
 kea                           x86_64    2.6.4-1.el9         epel         1.3 M
Installing dependencies:
 kea-libs                      x86_64    2.6.4-1.el9         epel         3.0 M
 log4cplus                     x86_64    2.0.5-15.el9        epel         337 k
 mariadb-connector-c           x86_64    3.2.6-1.el9_0       appstream    195 k
 mariadb-connector-c-config    noarch    3.2.6-1.el9_0       appstream    9.8 k
 postgresql-private-libs       x86_64    13.23-1.el9_7       appstream    140 k

Transaction Summary
================================================================================
Install  6 Packages

Total download size: 5.0 M
Installed size: 18 M
Downloading Packages:
(1/6): log4cplus-2.0.5-15.el9.x86_64.rpm                     1.8 MB/s | 337 kB     00:00
(2/6): kea-libs-2.6.4-1.el9.x86_64.rpm                       4.9 MB/s | 3.0 MB     00:00
(3/6): mariadb-connector-c-3.2.6-1.el9_0.x86_64.rpm          443 kB/s | 195 kB     00:00
(4/6): kea-2.6.4-1.el9.x86_64.rpm                            1.9 MB/s | 1.3 MB     00:00
(5/6): mariadb-connector-c-config-3.2.6-1.el9_0.noarch.rpm    59 kB/s | 9.8 kB     00:00
(6/6): postgresql-private-libs-13.23-1.el9_7.x86_64.rpm      868 kB/s | 140 kB     00:00
--------------------------------------------------------------------------------
Total                                                         42 kB/s | 5.0 MB     02:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                          1/1
  Installing       : postgresql-private-libs-13.23-1.el9_7.x86_64             1/6
  Installing       : mariadb-connector-c-config-3.2.6-1.el9_0.noarch          2/6
  Installing       : mariadb-connector-c-3.2.6-1.el9_0.x86_64                 3/6
```

Open ▾ kea-dhcp4 /etc/kea Save

```
 6 "data": "srv.world"
 7 },
 8 на описание
 9 { "code": 15,
10 "data": "user.net"
11 },
12 {
13 "name": "domain-search",
14 "data": "user.net"
15 }
16 { "name": "domain-name-servers",
17 "data": "192.0.2.1, 192.0.2.2"
18 },
19 на блок
20 { "name": "domain-name-servers",
21 "data": "192.168.1.1"
22 },
23 "subnet4": [
24 {
25 "id": 1,
26 // specify subnet that DHCP is used
27 "subnet": "192.168.1.0/24",
28 // specify the range of IP addresses to be leased
29 "pools": [ { "pool": "192.168.1.30 - 192.168.1.199" } ],
30 "option-data": [
31 {
32 // specify your gateway
33 "name": "routers",
34 "data": "192.168.1.1"
35 }
36 ]
37 }
38 ],
39
40 "interfaces-config": {
41 "interfaces": [ "eth1" ]
42 },
```

Plain Text ▾ Tab Width: 8 ▾ Ln 26, Col 36 ▾ INS

```
[   44.621644] vmwgfx 0000:00:02.0: [drm] DMA map mode: Caching DMA mappings.
[   44.624452] vmwgfx 0000:00:02.0: [drm] Legacy memory limits: VRAM = 262144 kB
, FIFO = 2048 kB, surface = 262144 kB
[   44.624462] vmwgfx 0000:00:02.0: [drm] MOB limits: max mob size = 131072 kB,
max mob pages = 262144
[   44.624467] vmwgfx 0000:00:02.0: [drm] Max GMR ids is 8192
[   44.624470] vmwgfx 0000:00:02.0: [drm] Max number of GMR pages is 1048576
[   44.624472] vmwgfx 0000:00:02.0: [drm] Maximum display memory size is 262144
kiB
[   44.743257] vmwgfx 0000:00:02.0: [drm] Screen Target display unit initialized
[   44.749736] vmwgfx 0000:00:02.0: [drm] Fifo max 0x00200000 min 0x00001000 cap
 0x00000355
[   44.750680] vmwgfx 0000:00:02.0: [drm] Using command buffers with DMA pool.
[   44.750704] vmwgfx 0000:00:02.0: [drm] Available shader model: SM_5.
[   44.754230] [drm] Initialized vmwgfx 2.20.0 20211206 for 0000:00:02.0 on mino
r 0
[   44.770198] fbcon: vmwgfxdrmfb (fb0) is primary device
[   44.786079] Console: switching to colour frame buffer device 160x50
[   44.800055] vmwgfx 0000:00:02.0: [drm] fb0: vmwgfxdrmfb frame buffer device
[   44.813079] snd_intel8x0 0000:00:05.0: intel8x0_measure_ac97_clock: measured 56364 usecs (2640 samples)
[   44.813091] snd_intel8x0 0000:00:05.0: clocking to 49190
[   50.307840] systemd-rc-local-generator[739]: /etc/rc.d/rc.local is not marked executable, skipping.
[   53.185485] NET: Registered PF_QIPCRTR protocol family
[   58.765459] e1000: eth0 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: RX
[   58.808401] IPv6: ADDRCONF(NETDEV_CHANGE): eth0: link becomes ready
[   58.900760] e1000: eth1 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: RX
[   58.965799] Warning: Unmaintained driver is detected: ip_set
[   59.814644] IPv6: ADDRCONF(NETDEV_CHANGE): eth1: link becomes ready
[   91.092689] block sda: the capability attribute has been deprecated.
```

Open Save

*kea-dhcp4.conf
/etc/kea

*kea-dhcp4.conf | *kea-dhcp4.conf

```
134         // When specifying options, you typically need to specify
135         // one of (name or code) and data. The full option specification
136         // covers name, code, space, csv-format and data.
137         // space defaults to "dhcp4" which is usually correct, unless you
138         // use encapsulate options. csv-format defaults to "true", so
139         // this is also correct, unless you want to specify the whole
140         // option value as long hex string. For example, to specify
141         // domain-name-servers you could do this:
142         // {
143         //     "name": "domain-name-servers",
144         //     "code": 15,
145         //     "csv-format": "true",
146         //     "space": "dhcp4",
147         //     "data": "user.net"
148         // }
149         // but it's a lot of writing, so it's easier to do this instead:
150         {
151             "name": "domain-name-servers",
152             "data": "192.168.1.1"
153         },
154
155         // Typically people prefer to refer to options by their names, so they
156         // don't need to remember the code names. However, some people likesS
157         // to use numerical values. For example, option "domain-name" uses
158         // option code 15, so you can reference to it either by
159         // "name": "domain-name" or "code": 15.
160         {
161             "code": 15,
162             "data": "example.org"
163         },
164
165         // Domain search is also a popular option. It tells the client to
166         // attempt to resolve names within those specified domains. For
167         // example, name "foo" would be attempted to be resolved as
168         // foo.mydomain.example.com and if it fails, then as foo.example.com
```

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```
"subnet4": [
    {
        // This defines the whole subnet. Kea will use this information to
        // determine where the clients are connected. This is the whole
        // subnet in your network.

        // Subnet identifier should be unique for each subnet.
        "id": 1,

        // This is mandatory parameter for each subnet.
        "subnet": "192.168.1.0/24",

        // Pools define the actual part of your subnet that is governed
        // by Kea. Technically this is optional parameter, but it's
        // almost always needed for DHCP to do its job. If you omit it,
        // clients won't be able to get addresses, unless there are
        // host reservations defined for them.
        "pools": [ { "pool": "192.168.1.30 - 192.168.1.199"  } ],

        // These are options that are subnet specific. In most cases,
        // you need to define at least routers option, as without this
        // option your clients will not be able to reach their default
        // gateway and will not have Internet connectivity.
        "option-data": [
            {
                // For each IPv4 subnet you most likely need to specify at
                // least one router.
                "name": "routers",
                "data": "192.168.1.1"
            }
        ],

        // Kea offers host reservations mechanism. Kea supports reservations
        // by several different types of identifiers: hw-address
        // (hardware/MAC address of the client), duid (DUID inserted by the
        // client), client-id (client identifier inserted by the client) and
        // circuit-id (circuit identifier inserted by the relay agent).
        //
```

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Остальные примеры задания конфигураций подсетей удалите.

3. Настройте привязку dhcpd к интерфейсу eth1 виртуальной машины server:


```
"interfaces-config": {
  "interfaces": [ "eth1" ]
},
```

4. Проверьте правильность конфигурационного файла:
   kea-dhcp4 -t /etc/kea/kea-dhcp4.conf

```
## Server configuration
config.vm.define "server", autostart: false do |server|
    server.vm.box = "rocky9"
    server.vm.hostname = 'server'
    server.vm.synced_folder ".", "/vagrant", disabled: true

    server.vm.boot_timeout = 1440

    server.ssh.insert_key = false
    server.ssh.username = 'vagrant'
    server.ssh.password = 'vagrant'

    server.vm.network :private_network,
            ip: "192.168.1.1",
                virtualbox__intnet: true

    server.vm.provision "server dummy",
                type: "shell",
                preserve_order: true,
                path: "provision/server/01-dummy.sh"
    server.vm.provision "server dns"
```

```
// DHCPv4 configuration starts here. This section will be read by DHCPv4 server
// and will be ignored by other components.
"Dhcp4": {
    // Add names of your network interfaces to listen on.
    "interfaces-config": {
        // See section 8.2.4 for more details. You probably want to add just
        // interface name (e.g. "eth0" or specific IPv4 address on that
        // interface name (e.g. "eth0/192.0.2.1").
        "interfaces": ["eth1" ]

        // Kea DHCPv4 server by default listens using raw sockets. This ensures
        // all packets, including those sent by directly connected clients
        // that don't have IPv4 address yet, are received. However, if your
        // traffic is always relayed, it is often better to use regular
        // UDP sockets. If you want to do that, uncomment this line:
        // "dhcp-socket-type": "udp"
    },

    // Kea supports control channel, which is a way to receive management
    // commands while the server is running. This is a Unix domain socket that
    // receives commands formatted in JSON, e.g. config-set (which sets new
    // configuration), config-reload (which tells Kea to reload its
    // configuration from file), statistic-get (to retrieve statistics) and many
    // more. For detailed description, see Sections 8.8, 16 and 15.
    "control-socket": {
        "socket-type": "unix",
        "socket-name": "kea4-ctrl-socket"
    },

    // Use Memfile lease database backend to store leases in a CSV file.
    // Depending on how Kea was compiled, it may also support SQL databases
```

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```
[root@server ~]# kea-dhcp4 -t /etc/kea/kea-dhcp4.conf
2026-03-04 19:52:51.138 INFO  [kea-dhcp4.hosts/27065.139698838464640] HOSTS_BACKENDS_REGISTER
ED the following host backend types are available: mysql postgresql
2026-03-04 19:52:51.144 WARN  [kea-dhcp4.dhcpsrv/27065.139698838464640] DHCPSRV_MT_DISABLED_Q
UEUE_CONTROL disabling dhcp queue control when multi-threading is enabled.
2026-03-04 19:52:51.145 WARN  [kea-dhcp4.dhcp4/27065.139698838464640] DHCP4_RESERVATIONS_LOOK
UP_FIRST_ENABLED Multi-threading is enabled and host reservations lookup is always performed
first.
2026-03-04 19:52:51.146 INFO  [kea-dhcp4.dhcpsrv/27065.139698838464640] DHCPSRV_CFGMGR_NEW_SU
BNET4 a new subnet has been added to configuration: 192.168.1.0/24 with params: t1=900, t2=18
00, valid-lifetime=3600
2026-03-04 19:52:51.146 ERROR [kea-dhcp4.dhcp4/27065.139698838464640] DHCP4_PARSER_FAIL faile
d to create or run parser for configuration element subnet4: specified reservation '192.0.2.2
01' is not within the IPv4 subnet '192.168.1.0/24'
Error encountered: specified reservation '192.0.2.201' is not within the IPv4 subnet '192.168
.1.0/24'
[root@server ~]# systemctl --system daemon-reload
[root@server ~]# systemctl enable kea-dhcp4.service
Created symlink /etc/systemd/system/multi-user.target.wants/kea-dhcp4.service → /usr/lib/syst
emd/system/kea-dhcp4.service.
[root@server ~]#
```

```
(gedit:27131): dconf-WARNING **: 19:54:00.846: failed to commit changes to dconf: Failed to e
xecute child process “dbus-launch” (No such file or directory)
[root@server ~]#  /var/named/master/fz
-bash: /var/named/master/fz: No such file or directory
[root@server ~]# cd  ..
[root@server /]# cd  /var/named/master/fz
-bash: cd: /var/named/master/fz: No such file or directory
[root@server /]# cd /var
[root@server var]# cd master
-bash: cd: master: No such file or directory
[root@server var]# systemctl restart named
Failed to restart named.service: Unit named.service not found.
[root@server var]# ping dhcp.user.net
ping: dhcp.user.net: Name or service not known
[root@server var]# chown -R named:named /var/named
chown: invalid user: ‘named:named’
[root@server var]#
```

```
root@server:/var

cockpit dhcpv6-client imap imaps pop3 pop3s smtp ssh
[root@server var]# firewall-cmd --get-services
firewall-cmd --add-service=dhcp
firewall-cmd --add-service=dhcp --permanent
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd a
udit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp
 bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ce
ph-mon cfengine checkmk-agent cockpit collectd condor-collector cratedb ctdb dds dds-multicas
t dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls docker-registry docker-swarm
dropbox-lansync elasticsearch etcd-client etcd-server finger foreman foreman-proxy freeipa-4
freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client gangli
a-master git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp ip
p-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kp
asswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-
controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-
scheduler-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvi
rt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memca
che minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula netbios-n
s netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storageconsol
e ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy promethe
us prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel
radius rdp redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-clie
nt samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap s
pideroak-lansync spotify-sync squid ssdp ssh steam-streaming svdrp svn syncthing syncthing-gu
i syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transm
ission-client upnp-client vdsm vnc-server warpinator wbem-http wbem-https wireguard ws-discov
ery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-c
lient xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
success
success
[root@server var]#
```

```
us prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel
radius rdp redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-clie
nt samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap s
pideroak-lansync spotify-sync squid ssdp ssh steam-streaming svdrp svn syncthing syncthing-gu
i syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transm
ission-client upnp-client vdsm vnc-server warpinator wbem-http wbem-https wireguard ws-discov
ery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-c
lient xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
success
success
[root@server var]# estorecon -vR /etc
bash: estorecon: command not found...
[root@server var]# restorecon -vR /etc
[root@server var]# restorecon -vR /var/named
restorecon: lstat(/var/named) failed: No such file or directory
[root@server var]# restorecon -vR /var/lib/kea/
[root@server var]# tail -f /var/log/messages
Mar  4 19:53:10 localhost systemd-rc-local-generator: /etc/rc.d/rc.local is not marked execut
able, skipping.
Mar  4 19:53:17 localhost systemd[1]: Reloading.
Mar  4 19:53:17 localhost systemd-rc-local-generator: /etc/rc.d/rc.local is not marked execut
able, skipping.
Mar  4 19:53:57 localhost gnome-shell[24477]: Window manager warning: Buggy client sent a _NE
T_ACTIVE_WINDOW message with a timestamp of 0 for 0x30000f8
Mar  4 19:53:57 localhost systemd[1]: Starting dnf makecache...
Mar  4 19:53:57 localhost dnf[27149]: Metadata cache refreshed recently.
Mar  4 19:53:58 localhost systemd[1]: dnf-makecache.service: Deactivated successfully.
Mar  4 19:53:58 localhost systemd[1]: Finished dnf makecache.
Mar  4 19:58:34 localhost systemd[1]: Starting PackageKit Daemon...
Mar  4 19:58:34 localhost systemd[1]: Started PackageKit Daemon.
```

Open *01-routing.sh Save

```
echo "Provisioning script $0"
nmcli connection modify "eth1" ipv4.gateway "192.168.1.1"
nmcli connection up "eth1"
nmcli connection modify eth0 ipv4.never-default true
nmcli connection modify eth0 ipv6.never-default true
nmcli connection down eth0
nmcli connection up eth0
```

```
    client.vm.provision "client routing",
                type: "shell",
                preserve_order: true,
                run: "always",
                path: "provision/client/01-routing.sh"

    client.vm.provider :virtualbox do |v|
        v.linked_clone = true
# Customize the amount of memory on the VM
        v.memory = 4000
```

```
xecute child process "dbus-launch" (No such file or directory)
[root@server /]# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fe69:a8d  prefixlen 64  scopeid 0x20<link>
        inet6 fd17:625c:f037:2:a00:27ff:fe69:a8d  prefixlen 64  scopeid 0x0<global>
        ether 08:00:27:69:0a:8d  txqueuelen 1000  (Ethernet)
        RX packets 1479236  bytes 2019630895 (1.8 GiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 178954  bytes 12543603 (11.9 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.1.1  netmask 255.255.255.0  broadcast 192.168.1.255
        inet6 fe80::a00:27ff:fee3:341f  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:e3:34:1f  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 44  bytes 5322 (5.1 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 17  bytes 2039 (1.9 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 17  bytes 2039 (1.9 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[root@server /]#
```

```
[root@server /]# cat /var/lib/kea/kea-leases4.csv
[root@server /]# mkdir -p /etc/named/keys
[root@server /]# mkdir -p /etc/named/keys
tsig-keygen -a HMAC-SHA512 DHCP_UPDATER > /etc/named/keys/dhcp_updater.key
[root@server /]# gedit etc/named/keys/dhcp_updater.key

(gedit:27344): dconf-WARNING **: 20:12:32.361: failed to commit changes to dconf: Fa
```

Open ▾ | dhcp_updater.key | /etc/named/keys | Save

```
key "DHCP_UPDATER" {
        algorithm hmac-sha512;
        secret "gy3Cy8Pe3eEsw7TmjEjJf1/K7RzFMSn5bw2HZ0XMhu7NUQNd1O17KDNI2ye5ZKJ1yytiWY15jKv/hwvi7ig0rg==";
};
```

```
** (gedit:27370): WARNING **: 20:14:42.249: Set document metadata failed: Setting attribute m
etadata::gedit-spell-language not supported

** (gedit:27370): WARNING **: 20:14:42.249: Set document metadata failed: Setting attribute m
etadata::gedit-encoding not supported

** (gedit:27370): WARNING **: 20:14:43.803: Set document metadata failed: Setting attribute m
etadata::gedit-position not supported

(gedit:27370): dconf-WARNING **: 20:14:43.832: failed to commit changes to dconf: Failed to e
xecute child process “dbus-launch” (No such file or directory)
[root@server /]# named-checkconf
bash: named-checkconf: command not found...
Packages providing this file are:
'bind'
'bind9.18'
[root@server /]#
```

Open ▾ | tsig-keys.json | /etc/kea | Save

```
"tsig-keys": [
{
"name": "DHCP_UPDATER",
"algorithm": "hmac-sha512",
"secret":
"psFFSGdUIwK36l1G82LOw15PuIH4W5uB8h/cw7F0XDsjniBbwm/59tM+V3ydcCs15VLpe2pxlUlg↪
}
],
```

Open ▾ kea-dhcp-ddns.conf /etc/kea Save

```
29   "forward-ddns" : {
30 "ddns-domains" : [
31 {
32 "name": "user.net.",
33 "key-name": "DHCP_UPDATER",
34 "dns-servers": [
35 { "ip-address": "192.168.1.1" }
36 ]
37 }
38 ]
39 },
40 "reverse-ddns" : {
41 "ddns-domains" : [
42 {
43 "name": "1.168.192.in-addr.arpa.",
44 "key-name": "DHCP_UPDATER",
45 "dns-servers": [
46 { "ip-address": "192.168.1.1" }
47 ]
48 }
49 ]
50 },
51 "loggers": [
52 {
53 "name": "kea-dhcp-ddns",
54 "output_options": [
55 {
56 "output": "stdout",
57 "pattern": "%-5p %m\n"
58 }
59 ],
60 "severity": "INFO",
61 "debuglevel": 0
62 }
63 ]
64 }
65 }
```

Saving file "/etc/kea/kea-dhcp-ddns.conf"... C ▾ Tab Width: 8 ▾ Ln 65, Col 2 ▾ INS

```
(gedit:27448): dconf-WARNING **: 20:18:21.001: failed to commit changes to dconf: Failed to e
xecute child process "dbus-launch" (No such file or directory)
[root@server /]# chown kea:kea /etc/kea/kea-dhcp-ddns.conf
[root@server /]# kea-dhcp-ddns -t /etc/kea/kea-dhcp-ddns.conf
Syntax check failed with: /etc/kea/tsig-keys.json:6.1: Invalid character: "
[root@server /]# systemctl enable --now kea-dhcp-ddns.service
Created symlink /etc/systemd/system/multi-user.target.wants/kea-dhcp-ddns.service → /usr/lib/
systemd/system/kea-dhcp-ddns.service.
[root@server /]# systemctl status kea-dhcp-ddns.service
× kea-dhcp-ddns.service - Kea DHCP-DDNS Server
     Loaded: loaded (/usr/lib/systemd/system/kea-dhcp-ddns.service; enabled; preset: disable>
     Active: failed (Result: exit-code) since Wed 2026-03-04 20:18:53 UTC; 14s ago
   Duration: 58ms
       Docs: man:kea-dhcp-ddns(8)
    Process: 27518 ExecStart=/usr/sbin/kea-dhcp-ddns -c /etc/kea/kea-dhcp-ddns.conf (code=ex>
   Main PID: 27518 (code=exited, status=1/FAILURE)
        CPU: 31ms

Mar 04 20:18:53 server.user.net systemd[1]: Started Kea DHCP-DDNS Server.
Mar 04 20:18:53 server.user.net kea-dhcp-ddns[27518]: 2026-03-04 20:18:53.091 INFO  [kea-dhc>
Mar 04 20:18:53 server.user.net kea-dhcp-ddns[27518]: 2026-03-04 20:18:53.095 FATAL [kea-dhc>
Mar 04 20:18:53 server.user.net kea-dhcp-ddns[27518]: Service failed: Could Not load configu>
Mar 04 20:18:53 server.user.net systemd[1]: kea-dhcp-ddns.service: Main process exited, code>
Mar 04 20:18:53 server.user.net systemd[1]: kea-dhcp-ddns.service: Failed with result 'exit->
lines 1-15/15 (END)
```

Open ▾ *kea-dhcp4.conf /etc/kea Save

```
// This is a basic configuration for the Kea DHCPv4 server. Subnet declarations
// are mostly commented out and no interfaces are listed. Therefore, the servers
// will not listen or respond to any queries.
// The basic configuration must be extended to specify interfaces on which
// the servers should listen. There are a number of example options defined.
// These probably don't make any sense in your network. Make sure you at least
// update the following, before running this example in your network:
// - change the network interface names
// - change the subnets to match your actual network
// - change the option values to match your network
//
// This is just a very basic configuration. Kea comes with large suite (over 30)
// of configuration examples and extensive Kea User's Guide. Please refer to
// those materials to get better understanding of what this software is able to
// do. Comments in this configuration file sometimes refer to sections for more
// details. These are section numbers in Kea User's Guide. The version matching
// your software should come with your Kea package, but it is also available
// in ISC's Knowledgebase (https://kea.readthedocs.io; the direct link for
// the stable version is https://kea.readthedocs.io/).
//
// This configuration file contains only DHCPv4 server's configuration.
// If configurations for other Kea services are also included in this file they
// are ignored by the DHCPv4 server.
{

"dhcp-ddns": {
"enable-updates": true
},
"ddns-qualifying-suffix": "user.net",
"ddns-override-client-update": true,

// DHCPv4 configuration starts here. This section will be read by DHCPv4 server
// and will be ignored by other components.
"Dhcp4": {
    // Add names of your network interfaces to listen on.
    "interfaces-config": {
        // See section 8.2.4 for more details. You probably want to add just
```

```
root@server:~
root@server:/var    root@server:/    root@server:~
tadata::gedit-spell-language not supported

* (gedit:27584): WARNING **: 20:20:34.091: Set document metadata failed: Setting attribute m
tadata::gedit-encoding not supported

* (gedit:27584): WARNING **: 20:20:34.772: Set document metadata failed: Setting attribute m
tadata::gedit-position not supported

gedit:27584): dconf-WARNING **: 20:20:34.791: failed to commit changes to dconf: Failed to e
ecute child process “dbus-launch” (No such file or directory)
root@server ~]# kea-dhcp4 -t /etc/kea/kea-dhcp4.conf
yntax check failed with: /etc/kea/kea-dhcp4.conf:26.1-11: syntax error, unexpected constant
tring, expecting Dhcp4
root@server ~]# systemctl restart kea-dhcp4.service
root@server ~]# systemctl status kea-dhcp4.service
 kea-dhcp4.service - Kea DHCPv4 Server
     Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4.service; enabled; preset: disabled)
     Active: failed (Result: exit-code) since Wed 2026-03-04 20:21:00 UTC; 8s ago
   Duration: 160ms
       Docs: man:kea-dhcp4(8)
    Process: 27628 ExecStart=/usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf (code=exited, st>
   Main PID: 27628 (code=exited, status=1/FAILURE)
        CPU: 61ms

ar 04 20:21:00 server.user.net systemd[1]: Started Kea DHCPv4 Server.
ar 04 20:21:00 server.user.net kea-dhcp4[27628]: 2026-03-04 20:21:00.871 INFO  [kea-dhcp4.d>
ar 04 20:21:00 server.user.net kea-dhcp4[27628]: 2026-03-04 20:21:00.873 ERROR [kea-dhcp4.d>
ar 04 20:21:00 server.user.net kea-dhcp4[27628]: 2026-03-04 20:21:00.873 ERROR [kea-dhcp4.d>
ar 04 20:21:00 server.user.net systemd[1]: kea-dhcp4.service: Main process exited, code=exi>
ar 04 20:21:00 server.user.net systemd[1]: kea-dhcp4.service: Failed with result 'exit-code>
ines 1-15/15 (END)
```

Вывод:

В ходе выполнения лабораторной работы были приобретены практические навыки по установке и конфигурированию DHCP-сервера.

Ответы на контрольные вопросы:

1. В каких файлах хранятся настройки сетевых подключений? - **В наиболее популярных операционных системах, таких как Windows и Linux, настройки сетевых подключений хранятся в различных файлах:**

   В Windows, основные настройки сетевых подключений, такие как IP-адрес, маска подсети, шлюз, DNS-серверы и другие, хранятся в реестре. Однако, конфигурационные данные также могут быть сохранены в текстовых файлах, таких как ipconfig или в файле конфигурации подключения.

   В Linux, настройки сети обычно хранятся в текстовых файлах в директории /etc/network/ или /etc/sysconfig/network-scripts/.

2. За что отвечает протокол DHCP? - **Протокол DHCP (Dynamic Host Configuration Protocol) отвечает за автоматическое присвоение сетевых настроек устройствам в сети, таких как IP-адресов, маски подсети, шлюза, DNS-серверов и других параметров.**

3. Поясните принцип работы протокола DHCP. Какими сообщениями обмениваются клиент и сервер, используя протокол DHCP? - **Принцип работы протокола DHCP:**

   Discover (Обнаружение): Клиент отправляет в сеть запрос на обнаружение DHCP-сервера.

   Offer (Предложение): DHCP-сервер отвечает клиенту, предлагая ему конфигурацию сети.

   Request (Запрос): Клиент принимает предложение и отправляет запрос на использование предложенной конфигурации.

   Acknowledgment (Подтверждение): DHCP-сервер подтверждает клиенту, что предложенная конфигурация принята и может быть использована.

4. В каких файлах обычно находятся настройки DHCP-сервера? За что отвечает каждый из файлов? - **Настройки DHCP-сервера обычно хранятся в файлах конфигурации, таких как:**

В Linux: /etc/dhcp/dhcpd.conf

В Windows: %SystemRoot%\System32\dhcp\dhcpd.conf

Они содержат информацию о диапазонах IP-адресов, параметрах сети и других опциях DHCP.

5. Что такое DDNS? Для чего применяется DDNS? - **DDNS (Dynamic Domain Name System) - это система динамического доменного имени. Она используется для автоматического обновления записей DNS, когда IP-адрес узла изменяется. DDNS применяется, например, в домашних сетях, где IP-адреса часто изменяются посредством DHCP.**

6. Какую информацию можно получить, используя утилиту ifconfig? Приведите примеры с использованием различных опций. - **Утилита ifconfig используется для получения информации о сетевых интерфейсах.**

Примеры:

ifconfig: Показывает информацию обо всех активных сетевых интерфейсах.

ifconfig eth0: Показывает информацию о конкретном интерфейсе (в данном случае, eth0).

7. Какую информацию можно получить, используя утилиту ping? Приведите примеры с использованием различных опций. - **Утилита ping используется для проверки доступности узла в сети.**

Примеры:

ping google.com: Пингует домен google.com.

ping -c 4 192.168.1.1: Пингует IP-адрес 192.168.1.1 и отправляет 4 эхо-запроса.